

BILL OF MATERIALS

LoRa Receiver Side (MQTT Bridge) Circuit

Currency: Indian Rupee (₹) | Quantity basis: 1 unit (prototype)

#	Ref. Des.	Component	Description / Value	Package	Qty	Unit Cost (₹)	Total Cost (₹)
1. Reset System Section							
1	R9	Resistor	10 kΩ pull-up (VCC to switch node)	0603	1	1	1
2	SW1	SW_Push_Dual	Tactile push-button switch, reset line	SMD	1	11	11
3	C8	Ceramic Capacitor	100 nF, debounce	0603	1	1	1
2. ESP32 Host MCU Section							
4	U2	ESP32-DEVKIT-V1	ESP32-WROOM-32 dev board (host MCU, UART bridge to LoRa module)	Module	1	400	400
3. MCU / LoRa Section (RAK3172)							
5	U1	RAK3172-xx-B-SM-xl	STM32WLE5-based LoRa module (868 MHz)	SMD-32	1	980	980
6	C2, C5, C6	Capacitor	100 nF x2, 10 μF x1, decoupling	0603/0805	3	3	9
7	E2	ANT-868-PW-RA	868 MHz tilt/swivel 1/4-wave monopole whip antenna, 4 dBi, screw-mount	Screw-mount	1	310	310
8	J1*	SMA Bulkhead Connector	RF feed for E2 (assumed; not explicit on schematic)	Panel-mount	1	40	40
4. PCB & Mechanical							
9	—	Custom PCB	2-layer FR4 fabrication (per unit, low volume)	—	1	350	350
ESTIMATED GRAND TOTAL (per unit)							₹ 2,102

- Notes**
- 1. Unit costs are approximate, indicative retail/distributor prices in Indian Rupees (₹) for single-unit / prototype quantities, based on Indian and international distributor listings as of September 2026. Actual prices vary with vendor, package variant, and exchange rate.
 - 2. Prices exclude GST, shipping/freight, PCB assembly (SMT) labour, and enclosure/mechanical hardware not shown on the schematic.
 - 3. *J1 (SMA bulkhead connector) is not explicitly drawn on the schematic but is the typical hardware needed to mount a screw-type external antenna like the ANT-868-PW-RA to an enclosure; included here as a reasonable assumption — omit if E2 mounts directly to a PCB-edge SMA footprint.
 - 4. The RAK3172 module cost (and, at scale, the ESP32 board cost) drops significantly at volumes of 1000+ units — bulk/OEM pricing should be requested from distributors for production runs.
 - 5. This BOM covers only the LoRa Receiver / MQTT Bridge side circuit shown in the supplied schematic (ESP32 host + RAK3172 LoRa module + reset circuit + antenna) — it does not include any power supply / battery charging section, which was not present on this schematic page.